Chapter 14, Problem 75P

Problem

In an *RLC* circuit, $R = 20\ \Omega$, $L = 4$ H, and $C = 1$ F. The circuit is magnitude-scaled by 10 and frequency-scaled by 105. Calculate the new values of the elements.

Step-by-step solution

Step 1 of 1

$$R' = K_m R = 20\times 10 = 200\Omega$$

$$L' = \frac{K_m}{K_f}\omega = \frac{10}{10^5}(4) = 400\ \mu H_Z$$

$$C' = \frac{C}{K_m K_f} = \frac{1}{10\times 10^5} = 1\ \mu F$$